

Cangrelor — The IV Lightning Rod P2Y₁₂ Inhibitor

THE IV LIGHTNING ROD - Cangrelor

INSIDE A STADIUM'S EMERGENCY MEDICAL STATION, NEAR CATH LAB

ZONE 1 MECHANISM & SPEED

1 Cangrelor IV bag with lightning bolt: IV only, immediate onset within seconds

2 All P2Y₁₂ gates snap shut instantly: rapid universal platelet inhibition

3 Clock "1/2 3-5 min": extremely short half-life

4 Clock "OFFSET 1 HOUR": complete reversal within 1 hour of stopping

5 Plates waking up, gates reopening: fastest on/off switch in class

6 "30µg/kg bolus, 4µg/kg/min × ≥2h": dosing

7 Spring latch on gates: reversible binding (like ticagrelor), not a prodrug

ZONE 2 THE TRAFFIC LIGHT FOR TRANSITION RULES

GREEN LIGHT
TICAGRELOR → ANYTIME ✓

BOTH REVERSIBLE → CAN OVERLAP → SAFE

RED LIGHT
CLOPIDOGREL / PRASUGREL → ONLY AFTER CANGRELOR STOPS

IRREVERSIBLE DRUGS NEED A FREE RECEPTOR — CANGRELOR BLOCKS ALL GATES → IF GIVEN DURING INFUSION, THE DOSE IS WASTED

TICAGRELOR = GREEN LIGHT ANYTIME | CLOPIDOGREL/PRASUGREL = RED LIGHT UNTIL IV STOPS

ZONE 3 INDICATION & SIDE EFFECTS

CATH LAB URGENCY

PCI IN PATIENTS NOT PRE-TREATED WITH ORAL P2Y₁₂

LOGIC: Patient rushed in with MI → hasn't taken pills → cangrelor provides IMMEDIATE protection during procedure

STENT AT WORK

BLEEDING — main risk, but easy to manage: JUST STOP THE IV → effect gone in 1 hour

DYSPNEA — like its cousin ticagrelor

1. Cangrelor IV bag

WHAT YOU SEE

IV pole and bag labeled 'Cangrelor' with a lightning bolt.

WHAT IT ENCODES

Cangrelor is the only INTRAVENOUS P2Y12 inhibitor — a direct-acting, reversible ATP analog. IV route gives near-immediate onset within seconds, used during PCI when oral agents are too slow or cannot be given.

BOARD TRAP

Cangrelor is IV-only and directly active (not a prodrug) — its niche is immediate intra-procedural inhibition, unlike all oral P2Y12 agents.

2. Half-life 3–5 minutes

WHAT YOU SEE

Clock 'HALF-LIFE: 3–5 MINUTES' in Zone 1.

WHAT IT ENCODES

Cangrelor has an ultra-short plasma half-life of 3–5 minutes due to rapid dephosphorylation by plasma esterases. Platelet function normalizes within ~60 minutes of stopping the infusion.

BOARD TRAP

3–5 min half-life is cangrelor's signature — its effect is fully controllable and gone ~1 hour after stop, unlike oral agents lasting days.

3. Offset within 1 hour

WHAT YOU SEE

Note 'OFFSET: when it stops, all gates pop open within 1 hour'.

WHAT IT ENCODES

Because binding is reversible and clearance is esterase-driven (not hepatic/renal), full reversal of platelet inhibition occurs within ~1 hour of discontinuation — ideal bridge to surgery or oral agents.

BOARD TRAP

Cangrelor offset (~1 h) is independent of renal/hepatic function — a key advantage in unstable or pre-surgical patients.

4. Bolus 30 µg/kg

WHAT YOU SEE

'BOLUS: 30 µg/kg before PCI' on the traffic-light placard.

WHAT IT ENCODES

Dosing: 30 µg/kg IV bolus immediately before PCI, then a continuous infusion. The bolus achieves rapid, near-complete P2Y12 blockade for the procedure.

BOARD TRAP

Remember 30 µg/kg bolus THEN 4 µg/kg/min infusion — bolus precedes PCI, infusion runs through it.

5. Infusion 4 µg/kg/min

WHAT YOU SEE

'INFUSION: 4 µg/kg/min for >=2 HOURS'.

WHAT IT ENCODES

After the bolus, cangrelor runs at 4 µg/kg/min for at least 2 hours or the duration of PCI, whichever is longer, maintaining universal platelet inhibition during the intervention.

BOARD TRAP

Infusion is 4 µg/kg/min for at least 2 h or the duration of PCI — both the bolus and infusion are weight-based.

6. Universal platelet inhibition

WHAT YOU SEE

'RAPID UNIVERSAL PLATELET INHIBITION' — all P2Y12 gates snap shut.

WHAT IT ENCODES

Cangrelor produces immediate, profound, and consistent P2Y12 blockade across all patients — no inter-individual variability from CYP genotypes because it is directly active and not metabolized to its active form.

BOARD TRAP

No CYP2C19 'poor metabolizer' problem with cangrelor — it sidesteps the clopidogrel variability entirely.

7. Uniquely controllable

WHAT YOU SEE

Dial labeled 'UNIQUELY CONTROLLABLE'.

WHAT IT ENCODES

Titratable on/off control is cangrelor's defining feature: start it for instant effect, stop it for full reversal in an hour — making it the only true 'on/off switch' antiplatelet.

BOARD TRAP

Controllability, not potency, is the cangrelor exam point — it bridges patients who cannot take or absorb oral P2Y12 agents.

8. Green light — ticagrelor anytime

WHAT YOU SEE

Traffic light GREEN: 'TICAGRELOR ANYTIME' — both reversible, overlap is safe.

WHAT IT ENCODES

Transition rule: ticagrelor (also reversible) can be started ANYTIME during or after the cangrelor infusion — no receptor competition problem because neither irreversibly occupies the site permanently.

BOARD TRAP

Ticagrelor = green light = give anytime with cangrelor; this contrasts sharply with the thienopyridine red-light rule below.

9. Red light — clopidogrel/prasugrel after stop

WHAT YOU SEE

Traffic light RED: 'CLOPIDOGREL / PRASUGREL — only AFTER cangrelor stops'.

WHAT IT ENCODES

Critical interaction: the active metabolites of clopidogrel and prasugrel cannot bind P2Y12 while cangrelor occupies it, so they are WASTED if given during the infusion. Give them only after cangrelor is stopped (or at end of infusion).

BOARD TRAP

Irreversible thienopyridines must wait until cangrelor stops — dosing them during infusion leaves the patient unprotected once cangrelor clears.

10. P2Y12 gates snap shut

WHAT YOU SEE

Row of P2Y12 gates labeled 'ALL P2Y12 GATES SNAP SHUT'.

WHAT IT ENCODES

Cangrelor reversibly occupies essentially all P2Y12 receptors during infusion, blocking ADP-mediated platelet activation and aggregation immediately and completely.

BOARD TRAP

Because it physically occupies the receptor, thienopyridine active metabolites have nowhere to bind — the mechanistic basis of the red-light rule.

11. Cath lab urgency / ambulance

WHAT YOU SEE

Ambulance 'AMBULANCE PULLS UP TO THE STADIUM, PT RUSHIN' INTO CATH LAB'; 'CARDIOLOGIST'.

WHAT IT ENCODES

Indication: PCI in P2Y12-naïve patients not pre-treated with an oral agent. Logic — patient rushed in with MI hasn't taken pills, so cangrelor gives immediate intra-procedural protection.

BOARD TRAP

Cangrelor's indication is the NOT-pretreated PCI patient — if oral P2Y12 was already loaded and absorbed, cangrelor adds little.

12. Stent at work

WHAT YOU SEE

'STENT AT WORK' graphic near the cardiologist.

WHAT IT ENCODES

During stent deployment, immediate platelet inhibition prevents acute stent thrombosis. Cangrelor (CHAMPION PHOENIX trial) reduced peri-procedural ischemic events vs clopidogrel in PCI.

BOARD TRAP

Cangrelor bridges the gap until oral agents take effect — its benefit is peri-procedural, then you transition to an oral P2Y12 inhibitor.

13. Bleeding — main risk

WHAT YOU SEE

Band-aids: 'BLEEDING — main risk, but easy to manage: just stop the IV; effect gone in 1 hour'.

WHAT IT ENCODES

As with all potent antiplatelets, bleeding is the principal adverse effect, but cangrelor's ultra-short half-life makes it uniquely manageable: simply stopping the infusion restores hemostasis within ~1 hour.

BOARD TRAP

There is no specific reversal agent needed for cangrelor — stopping the drug IS the reversal, thanks to the 3–5 min half-life.

14. Dyspnea (class cousin)

WHAT YOU SEE

Dyspnea balloon: 'shared side effect with cousin ticagrelor'.

WHAT IT ENCODES

Like ticagrelor, cangrelor can cause transient dyspnea (adenosine-related class effect of direct-acting reversible P2Y12 agents). Usually mild and resolves as infusion stops.

BOARD TRAP

Dyspnea links cangrelor and ticagrelor — both directly acting reversible agents — and is absent with the prodrug thienopyridines.
